Results

01 · ARIMA / SARIMA

model & forecast one series

Table 1. Model summary

	(1)
sar1	−0.48
	(0.12)
	[−0.71, −0.25]
drift	1.67
	(0.04)
	[1.60, 1.74]
Num.Obs.	72
σ^2	21.66
Ljung–Box p	<.001
AIC	361.95
BIC	368.23
Log.Lik.	−177.97

Ljung–Box Q(23) = 169.43, p < .001 (lag 24, residual autocorrelation)

arima()/auto.arima() return each coefficient with its SE (CI via confint()); they do not produce per-coefficient z/p — add lmtest::coeftest() if those are wanted. The Ljung–Box row tests residual autocorrelation jointly to lag = max(10, 2 × seasonal period), with df adjusted for the number of fitted ARMA terms (p + q + P + Q).

Table 2. Forecast

Period	Forecast	80% PI	95% PI
1	226.60	[220.64, 232.56]	[217.48, 235.72]
2	236.15	[230.19, 242.12]	[227.03, 245.27]
3	241.39	[235.43, 247.36]	[232.27, 250.51]
4	244.45	[238.49, 250.42]	[235.33, 253.57]
5	252.00	[246.03, 257.96]	[242.88, 261.12]
6	248.34	[242.38, 254.30]	[239.22, 257.46]
7	242.50	[236.53, 248.46]	[233.38, 251.62]
8	240.04	[234.08, 246.01]	[230.92, 249.17]
9	234.35	[228.39, 240.32]	[225.23, 243.47]
10	234.30	[228.34, 240.26]	[225.18, 243.42]
11	234.14	[228.18, 240.11]	[225.02, 243.27]
12	237.35	[231.39, 243.32]	[228.23, 246.47]

Figure. Forecast

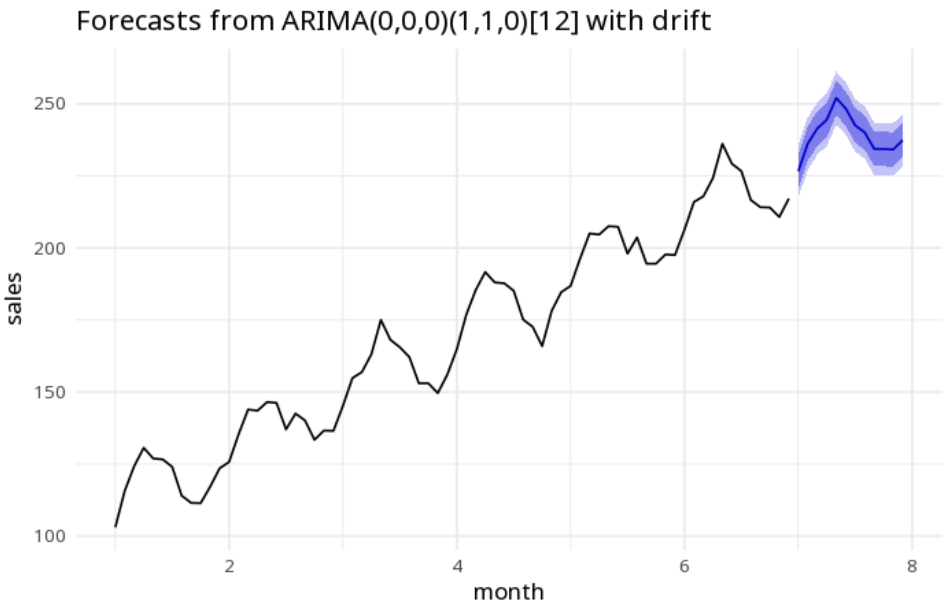
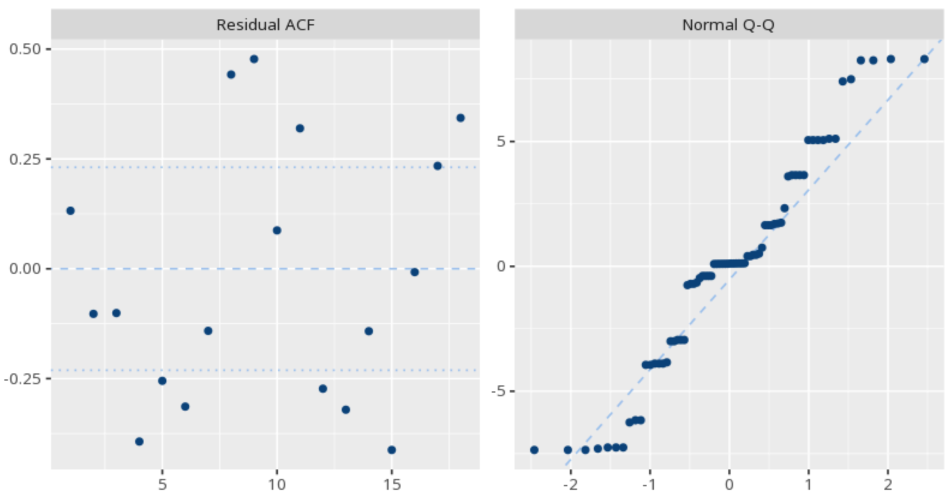


Figure. Residual diagnostics



How to read this test

The chosen (p,d,q)(P,D,Q) orders describe the model. The Ljung–Box Q(df) row tests whether the residuals still carry autocorrelation up to the reported lag (a large p means none is detected); use it with AIC/BIC to judge fit. The forecast table and plot give predictions with uncertainty bands. AIC/BIC compare models only when fit to the same series with identical differencing (d, D) and transformation; lower is better, but they are relative, not absolute, measures of fit. Method: R forecast package (Hyndman & Khandakar, 2008; Hyndman & Athanasopoulos, Forecasting: Principles and Practice).

APA template: An ARIMA(0,0,0)(1,1,0[12]) model was fit (AIC=361.95); the Ljung–Box test of residual autocorrelation gave $p < .001$.